

Due on tuesday, April 25th, in class

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1. As measured in frame Σ , the coordinates of events A and B are respectively (ct_A, x_A) and (ct_B, x_B) . The interval between the events are spacelike.

There is then a frame Σ' in which the two events are simultaneous.

- (a) [4 pts] Plot the two events on a spacetime diagram, taking the Σ frame to have ct and x axes perpendicular to each other. Plot also the axes of the Σ' frame, so that it is clear that A and B are simultaneous in this frame. You only get credits if this last feature is evident in your drawing.
 - (b) [4 pts] Using the spacetime diagram, calculate the speed v of the Σ' frame relative to the Σ frame, in terms of t_A, t_B, c, x_A, x_B .
 - (c) [3 pts] Now calculate the same expression for v using the Lorentz transformations.
2. [4 pts] The 4-velocity and 4-acceleration of a particle are given by

$$V = (\gamma_v c, \gamma_v \vec{v}), \quad A = \left(\frac{\gamma_v^4}{c} v \dot{v}, \frac{\gamma_v^4}{c^2} v \dot{v} \vec{v} + \gamma_v^2 \vec{a} \right),$$

where $\vec{v}$ is the 3-velocity, $v = \sqrt{\vec{v} \cdot \vec{v}}$ is the speed, and $\dot{v} = \frac{dv}{dt}$ is NOT the 3-acceleration $\vec{a}$. Show that the 4-velocity and 4-acceleration are ‘orthogonal’ to each other, i.e., have zero inner product.

From problem set 1, we know $v \dot{v} = \vec{v} \cdot \vec{a}$.

3. The four-vector A^μ is found to be timelike in one inertial frame.
 - (a) [3 pts] Write an inequality expressing the fact that A^μ is timelike.
 - (b) [3 pts] Explain why A^μ has to be timelike in any inertial frame.
4. [3 pts] The mass m and the charge q of a particle are four-scalars. Explain why the combination

$$(m, q, m, q)$$

is not a four-vector.

5. Four-momenta.

- (a) [4 pts] Show that the four-momentum of a particle with nonzero mass is always timelike.
- (b) [3 pts] Write down the four-momentum of a photon moving in the y direction. The frequency of the photon is denoted by f .
- (c) [4 pts] Find out whether the 4-momentum of a photon is timelike, spacelike, or lightlike/null.

6. [6 pts] A π -meson (a.k.a. pion) of mass M at rest disintegrates into a μ -meson (a.k.a. muon) of mass m_μ and a neutrino of effectively zero mass. Show that the kinetic energy of motion of the muon is

$$T = \frac{(M - m_\mu)^2}{2M} c^2$$

As usual: first draw the situations before and after the reaction/decay, write down equations for energy and momentum conservation in terms of the most useful variables, and then manipulate these equations as required.

You should decide early whether it is helpful to work with speed and γ , or whether to avoid these and work with momentum instead.

You probably want to use $c = 1$ units for this problem.

7. Electromagnetism.

- (a) [4 pts] Show that the tensor equation $\partial_\mu J^\mu = 0$ is equivalent to the continuity equation of electromagnetism. Here J^μ is the current density 4-vector.
- (b) [5 pts] The two inhomogeneous Maxwell's equations can be expressed as the tensor equation for the field tensor $F^{\mu\nu}$:

$$\partial_\mu F^{\mu\alpha} = J^\alpha$$

Derive the continuity equation from this equation, using the fact that the field tensor is antisymmetric.

Note: Depending on the definition conventions for $F^{\mu\nu}$ and J^μ , Maxwell's inhomogeneous equations may also appear in textbooks in the forms

$$\partial_\mu F^{\mu\alpha} = -J^\alpha \quad \text{or} \quad \partial_\mu F^{\mu\alpha} = \mu_0 J^\alpha \quad \text{or} \quad \partial_\mu F^{\mu\alpha} = -\mu_0 J^\alpha$$

but this will not matter for the problem above.